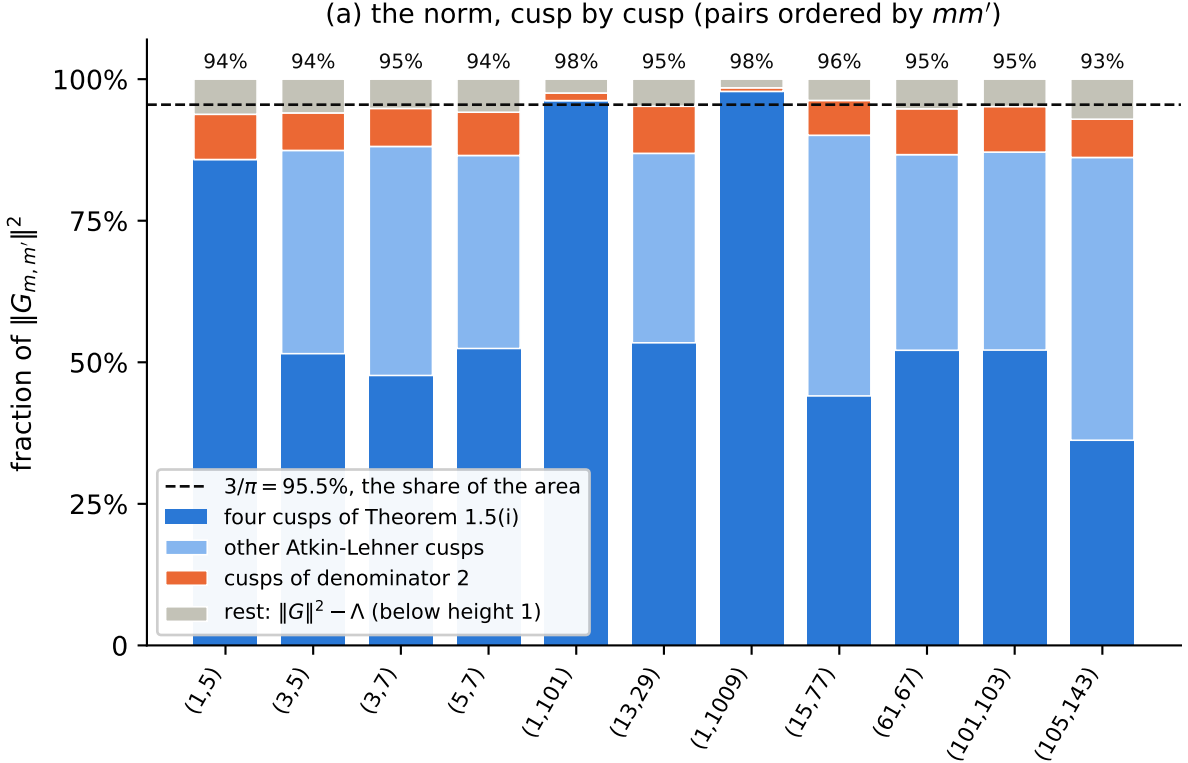
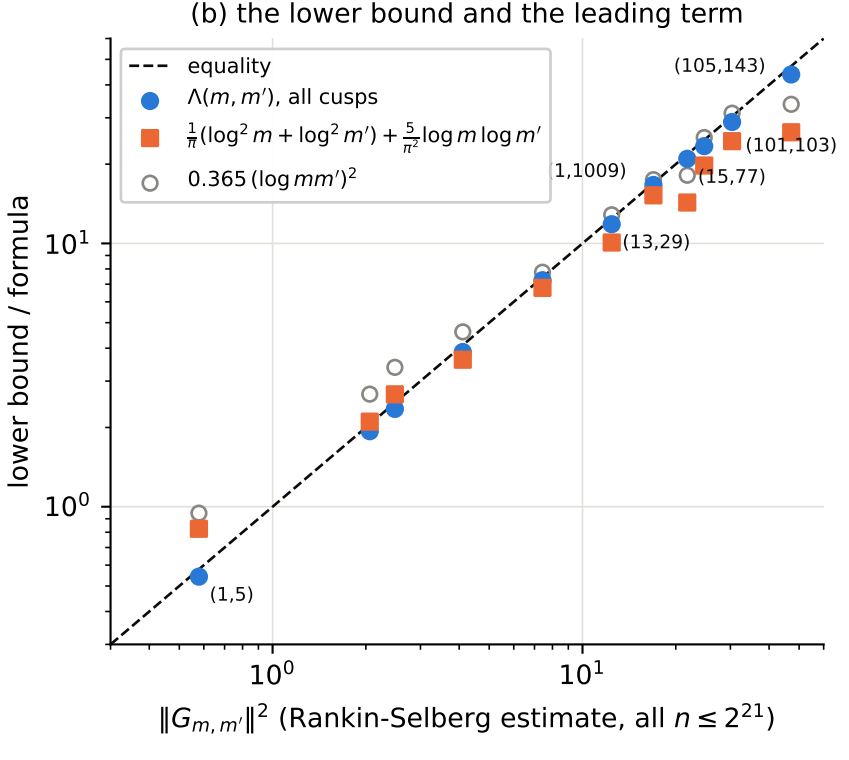

(a) the norm, cusp by cusp (pairs ordered by mm')
fraction of $\|G_{m,m'}\|^2$
100%
75%
50%
25%
0
94%
94%
95%
94%
98%
95%
98%
96%
95%
95%
93%
$3/\pi = 95.5\%$, the share of the area
four cusps of Theorem 1.5(i)
other Atkin-Lehner cusps
cusps of denominator 2
rest: $\|G\|^2 - \Lambda$ (below height 1)
(1,5)
(3,5)
(3,7)
(5,7)
(1,101)
(13,29)
(1,1009)
(15,77)
(61,67)
(101,103)
(105,143)


(b) the lower bound and the leading term
lower bound / formula
equality
$\Lambda(m,m')$, all cusps
$\frac{1}{\pi}(\log^2 m + \log^2 m') + \frac{5}{\pi^2}\log m \log m'$
$0.365\,(\log mm')^2$
(105,143)
(101,103)
(15,77)
(1,1009)
(13,29)
(1,5)
10^1
10^0
10^0
10^1
$\|G_{m,m'}\|^2$ (Rankin-Selberg estimate, all $n \le 2^{21}$)